

Numerical investigation of flow and heat transfer in a volumetric solar receiver



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ABSTRACT

Volumetric solar receivers are used in solar power plants to convert concentrated solar radiation into high temperature heat to operate a thermal engine. In general, porous high temperature materials are used for this purpose. Since the pore geometry is important for the efficiency performance of the receiver, current R&D activities focus on the optimization of this quantity. In this study, the influence of slight geometry changes of this component on its temperature distribution and efficiency has been investigated with the objective of an overall improvement. A numerical analysis of the mass and heat transfer through the receiver has been performed. **The investigated receiver was an extruded honeycomb structure made out of Silicon Carbide.** Additionally, experimental tests have been performed. In these tests, selected receiver samples have been exposed to concentrated radiation. From these tests solar-to-thermal efficiency data have been derived, which could be compared with the calculated data. Two numerical models have been developed. One makes use of the real geometry of the channel (single channel model), the other one considers the receiver to be “porous continuum”, which is described by homogenized properties such as permeability and effective heat conductivity. The experimental parameters such as the average solar heat flux and the mass flow were taken into account in the models as boundary conditions. Various quantities such as the average air outlet temperatures, the temperature distributions and the solar-to-thermal efficiency were used for the comparison. The correspondence between the experimental and numerical results of both numerical models confirms the capability of the approaches for further studies.

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1. Introduction

The volumetric solar receiver is one of the key components of *solar tower technology*, which uses concentrated solar radiation to generate thermal energy to operate a thermal engine. The concentrated radiation is generated by a large field of individually moving mirrors (heliostats). The focus is located on top of a tower (Fig. 1, left). The receiver, located in the focus, generally consists of a porous material, which absorbs the radiation and transfers it into heat of an air flow. After that, the air is used to generate heat in a boiler.

A reliable, continuous and predictable operation of the solar receiver is of great importance. It has been investigated already since 1985. From this time, numerous studies have been published [1–4]. A comprehensive review is given in Ref. [5]. The material investigated is an improved Silicon Carbide honeycomb structure

based on the state-of-the-art-technology shown in Fig. 1 (middle and right), which has also been applied in the solar tower in Jülich [6]. It has been manufactured by St. Gobain IndustrieKeramik Rödental.

In the current study, the performance of one single receiver module has been investigated experimentally and numerically. The numerical work was carried out using two different models of the volumetric solar receiver. The first of them, the so called “single channel model” describes air flow and heat transfer in a selected volume in detail and the second one (“continuum model”) describes air flow and heat transfer in the whole volume of the module treating the absorber as a porous medium described by effective properties.

In the numerical work presented, the influence of geometrical variations on the overall performance of the receiver was investigated with the objective to further increase the efficiency of this component. The geometry of the receiver modules can be seen in Fig. 1. They consist of a honeycomb-like ceramic structure made out of channels and walls named “absorber”: This absorber is embedded in a ceramic containment, the so-called “cup”. Two geometrical

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Nomenclature		z	coordinate along the main flow direction, m
A	inlet surface of the receiver, m^2	Greek symbols	
A_V	specific surface, m^2/m^3	α	convective heat transfer coefficient, $W/(m^2 K)$
C_p	specific isobaric heat, $J/(kg K)$	η	coefficient of the dynamical viscosity, $kg/(m s)$
C	emissivity · Stefan Boltzmann constant, $W/(m^2 K^4)$	ε_p	porosity, –
I	radiative flux density, W/m^2	ρ	density, kg/m^3
K	permeability, m^2	ξ	extinction coefficient of the radiation, $1/m$
k	thermal conductivity, $W/(m K)$	φ	inlet angle of the radiation flow, °
$\dot{m}$	mass flow rate density, $kg/(s m^2)$	ε	emissivity, –
$\dot{M}$	mass flow rate, kg/s	γ	reflectivity
Nu	Nusselt number, –	Indices	
p	pressure, Pa	rad	radiative
Q	power, W	abs	absorptive
q	power per unit volume, W/m^3	0	located at the irradiated front surface (radiative flux density)
q_0	solid/fluid convective heat exchange, W/m^3	inlet	located at the irradiated front surface (entrance of the air flow into the absorber)
Re	Reynolds number, –	outlet	located at the outlet of the air flow from the absorber
T	air temperature, K		
T_2	solid temperature, K		
u	velocity, m/s		

variations have been chosen for the investigations in this study. The first one considers two different channel diameter/wall thickness dimensions and the second one considers the overall shape of the absorber.

2. Methodology

2.1. Experimental set-up

To investigate performance characteristics of the receiver, a tubular experimental set-up was used, which was able to supply boundary conditions similar to the application in the power plant for one 140°140 mm receiver module. Fig. 2 shows a photograph of the whole set-up, Fig. 3 shows the flow scheme. The receiver cup is plugged into a high temperature steel tube, which was insulated with high-temperature-materials based on ceramic fibers. The concentrated radiation of up to 900 kW/m² is provided by 10 Xenon short-arc lamps with ellipsoid reflectors. As ellipses have two foci, the short arc lamps are located in the first focus as emitters, the sample is located in the second focus. The distance between the lamps and the sample is approximately 150 cm.

The concentrated radiation meets the inlet surface of the receiver sample and heats it up. The blower supplies the pressure difference and causes an air flow through the sample. Thus cold

ambient air is heated up to defined temperatures, which are measured with thermocouples at the locations shown in Fig. 3. At each location at least 3 thermocouples were used. The test bed can be used for tests with and without air return flow. For the purpose of the current study, no return air flow was used, because the capability of the material to convert radiation into heat should be tested. A counter flow air/water heat exchanger was used to cool down the air temperature to protect the following components (mass flow meter, blower) from high temperatures. Furthermore the determination of the absorbed energy is easier and more precise if the enthalpy gain of the water circuit is measured. Besides air and water temperatures also flow rates of water and air are measured and logged with a standard pc-based data acquisition system.

Additionally, radiation flux and surface temperature are measured with separate devices. For the radiation flux, the camera-target system FATMES with an accuracy of ±5% is employed [7]. Surface temperature has been monitored with standard infrared thermography.

Efficiency is calculated according to equation (1):

$$\eta = Q_{\text{abs}}/Q_{\text{rad}} \quad (1)$$

Here, Q_{rad} is the irradiative flux from the lamps penetrating onto the aperture of the sample. It has been determined by integrating

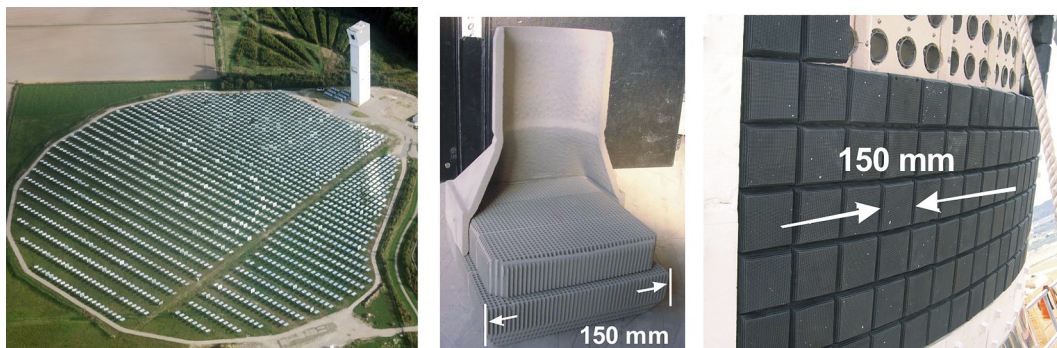


Fig. 1. Photographs of the Jülich Solar Tower (left), a part of a single absorber module and the solar receiver during installation.

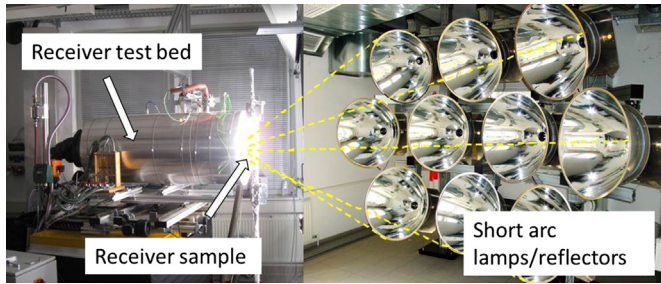


Fig. 2. Experimental set-up for absorber testing during operation (left), Xenon short-arc-lamps (right).

the flux density map, which has been generated with FATMES, Q_{abs} is the measured enthalpy difference in the water circuit and the enthalpy difference in the air flow behind the heat exchanger:

$$Q_{\text{abs}} = \dot{M}_{\text{inlet}} \cdot c_p \cdot (T_{\text{outlet}} - T_{\text{inlet}}) \quad (2)$$

Solar-to-thermal efficiency and air outlet temperature are the parameters, which have been used for the comparison of the different absorber types and the experimental and numerical results.

2.2. Numerical models

The two models developed are 3D-models and calculate the entering concentrated radiation as well as the air flow and heat transfer inside the porous receiver material. They have been developed with the COMSOL Multiphysics code. The first one calculates the flow inside the real geometry of the absorber's honeycomb-structure. Because of this complex structure, this method requires a high number of elements in the flow region. Thus, it may be applied only to a small excerpt of the total volume of the receiver element. It cannot be used to describe the thermal behavior of the whole receiver component consisting of more than 2000 channels. Therefore it is called the *single-channel-model*.

The second model considers the volume of the receiver as a porous continuum with homogenized properties, which have been either experimentally determined (permeability, effective heat conductivity) or have been derived from the single-channel-model. It is called the *continuum model*. It can be used to describe the thermal behavior of the larger components, such as in this case one-quarter of the whole receiver component consisting of absorber and cup. It is able to describe radial heat transfer, which occurs perpendicular to the main flow direction.

For the *single channel model* the following COMSOL modules were used: (1) weakly compressible Navier–Stokes, (2) convection and heat conduction in the air and (3) heat conduction in the solid phase. The differential equations referring to the modules are provided in Table 1. The Navier–Stokes module includes the Navier–Stokes equation, the energy conservation law and the equation of continuity. By simultaneously solving these equations the motion of a Newtonian fluid depending on the pressure at the boundary and viscous effects inside the fluid may be described. The other two modules use the general equations for conductive and convective heat transfer.

In case of the *continuum model* the Brinkmann-equation for porous materials instead of the Navier–Stokes equation has been used for domains, in which flow through a porous medium is described (Table 2).

Both, the single channel model as well as the continuum model make it possible to calculate velocity and temperature fields in the single channel and in the total volume of the receiver element. Since two heat transfer equations are solved simultaneously for the gaseous and the solid phase of the porous continuum two independent temperature fields are calculated. Energy exchange between the two phases is regarded by integrating a term describing the volumetric solid-to-fluid heat transfer by convection:

$$q_0 = \alpha \cdot A_V (T_2 - T)$$

Because of the full symmetry of the geometry in the single channel as well in the whole receiver module, in both models only one-quarter of the volume was calculated.

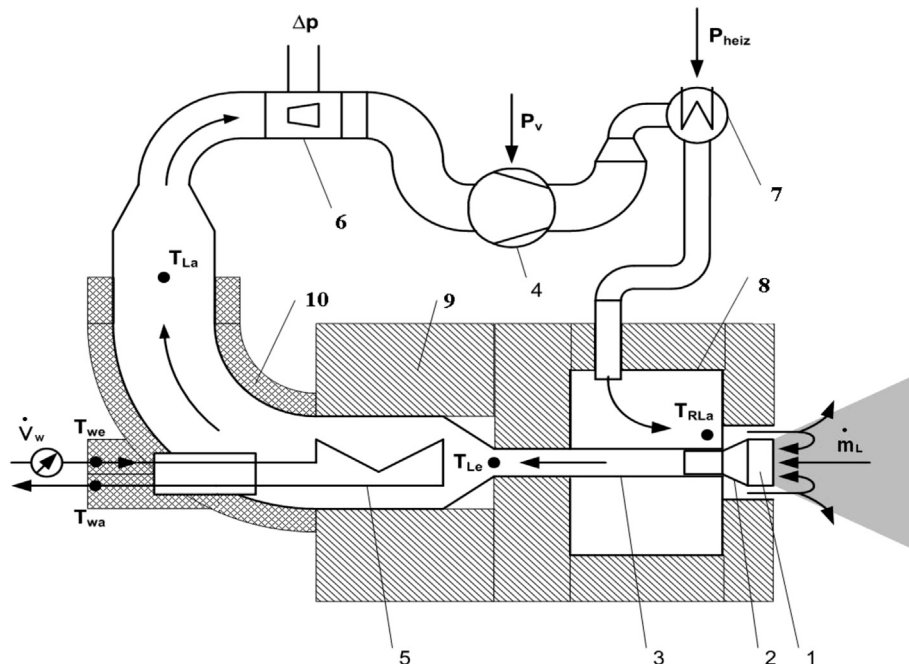


Fig. 3. Flow-scheme of the experimental set-up. 1 – absorber sample, 2 – receiver cup, 3 – hot gas tube, 4 – air blower, 5 – heat exchanger, 6 – mass flow measurement, 7 – heater for return air (not used), 8 – return-air chamber, 9/10 – thermal insulation.

Table 1
Area and boundary conditions of the single-channel model.

Module	Domain conditions	Boundary conditions	
		Inlet	Outlet
Weakly compressible Navier–Stokes	$\eta \cdot \nabla(\nabla u + (\nabla u)^T) = \rho \cdot u \cdot \nabla u$ $\nabla(\rho \cdot u) = 0 \quad \rho = \rho(p, T)$	Velocity $u = u_0$	Pressure without the viscose stress $p = p_0$
Convection and heat conduction in air	$\nabla(-k \cdot \nabla T + \rho \cdot c_p \cdot u \cdot T) = q_0$	Temperature $T = T_0$	Convection flow $n \cdot (-k \cdot \nabla T) = 0$
Heat conduction in the solid body	$-\nabla(k \cdot \nabla T_2) = q_0$ $q_0 = Q_{\text{absorb}} \cdot \frac{\xi \cdot \exp(-\xi \cdot z)}{A_0}$	Heat flow (radiation) $n \cdot (k \cdot \nabla T_2) = q_0 + C \cdot (T_{\text{amb}}^4 - T_2^4)$ $q_0 = I_0 - F \cdot (T_2 - T_0)$ $C = \varepsilon \cdot 5.67 \cdot 10^{-8}$	Temperature $T_2 = T$

2.2.1. Boundary and area conditions of the numerical models

The concept of a volumetric heat source of the absorbed heat in the solar receiver was used for the calculation. As it is known from previous experimental and numerical investigations, which are reviewed in Ref. [5], the intensity of the radiation is attenuated in the cellular material due to a multiple reflection/absorption process, it can be described by an exponential law:

$$I = I_0 \cdot e^{-\xi \cdot z} \quad (3)$$

Here, I_0 [W/m²] is the inlet heat flux, ξ [1/m] is the extinction coefficient and z is the coordinate in flow direction. The heat flux I_0 used for the simulation was according to the average experimental value in the solar simulator, which has been 650 kW/m². This value is according to the lay-out data of the Jülich Solar Tower at design point. The extinction coefficient was found experimentally (through transmittance measurements of the samples with different thickness) as it was described in Ref. [5]. It was 2300 1/m.

The main area- and boundary-conditions of both numeric models are presented below in Tables 1 and 2 as simplified descriptions.

A more detailed description of the continuum model is presented in Ref. [8]. The inlet value of the radiation flux with the extinction coefficient and volumetric heat transfer coefficient was used for the determination of the volumetric heat source for the air and for the solid body. The volumetric heat transfer coefficient $\alpha \cdot A_V$ was calculated from the specific surface A_V of the sample [m²/m³] and coefficient of the heat transfer, which was determined from the numerical heat flow distribution of the single channel model.

3. Results and discussions

The single channel model was used to compare two different channel/wall geometries of the receiver element as far as the influence on the receiver's solar-to-thermal efficiency is concerned. Both geometries have been also investigated experimentally. In contrast to this, a geometry modification of the whole receiver

module was investigated with the continuum model. An additional frustum-like geometry has been chosen by the manufacturer to homogenize the receiver front temperature. In the experimental tests and in the calculations this geometry was expected to show improved efficiency properties.

3.1. The results of the single channel model

The two different channel/wall dimensions considered are shown in Table 3 together with the corresponding values of the specific surface area and additional flow and radiation parameters. The values for mass flow, inlet temperature and absorbed radiation have been set according to the two experiments with samples with the dimensions mentioned. Note that the same radiation conditions have been chosen but since the second sample consists of smaller dimensions the calculated region was smaller and so also the absolute value of radiative power is smaller. Fig. 4 shows a schematic cross-sectional view of the two geometries considered.

This numeric calculation yields flow velocity, solid and fluid temperature as well as fluid density and pressure at any location of the domains under consideration. Fig. 5 shows the air temperature distribution in the channel with the standard geometry. Passing the channel the air heats up from 40° to more than 750° at the outlet. Fig. 6 shows the corresponding wall-to-fluid heat flow distribution along the channel in flow direction. The air temperature distribution shows large temperature gradients in the inlet area. After approximately 20 mm the temperature proceeds constantly. This has been expected due to the large difference between fluid and

Table 3
Sizes and inlet parameters of the cylindrical samples.

Sample	Channel width [m]	Wall thickness [m]	Specific surface [m ² /m ³]	Mass flow for the whole sample [kg/s]	T_{inlet} [°C]	Q_{abs} [W]	I_0 [kW/m ²]
1	0.00218	0.00055	1170	$2.4 \cdot 10^{-3}$	45.1	1.16	650
2	0.00143	0.00025	2027	$2.5 \cdot 10^{-3}$	47.0	0.43	650

Table 2
Area and boundary conditions of the continuum model.

Module	Domain conditions	Boundary conditions	
		Inlet	Outlet
Weakly compressible Navier–Stokes Brinkmann	$\eta \cdot \nabla(\nabla u + (\nabla u)^T) = \rho \cdot u \cdot \nabla u$ – in air $\nabla(\rho \cdot u) = 0 \quad \rho = \rho(p, T)$ $(\frac{1}{\varepsilon_p}) \eta \cdot \nabla(\nabla u + (\nabla u)^T) = (\frac{\eta}{K}) \cdot u$ – in porous material	Velocity $u = u_0$	Pressure without the viscose stress $p = p_0$
Convection and heat conduction in air	$\nabla(-k \cdot \nabla T + \rho \cdot c_p \cdot u \cdot T) = q_0$ $q_0 = \alpha \cdot A_V (T_2 - T)$	Temperature $T = T_0$	Convection flow $n \cdot (-k \cdot \nabla T) = 0$
Heat conduction in the solid body	$-\nabla(k \cdot \nabla T_2) = q_0$ $q_0 = I_0 \cdot \varepsilon_p \cdot \xi \cdot \exp(-\xi \cdot z) \cdot \cos \alpha + \alpha \cdot A_V \cdot (T - T_2)$ $k = (1 - \varepsilon_p) \cdot 280.7 \cdot \exp(-0.0021 \cdot T_2)$	Heat flow (radiation) $k \cdot \nabla T_2 = q_0 + C \cdot (T_{\text{amb}}^4 - T_2^4)$ $q_0 = I_0 \cdot \cos \varphi \cdot (1 - \varepsilon_p) - (1 - \varepsilon_p) \cdot F \cdot (T_2 - T_0)$ $C = \varepsilon \cdot 5.67 \cdot 10^{-8}$	Temperature $T_2 = T$

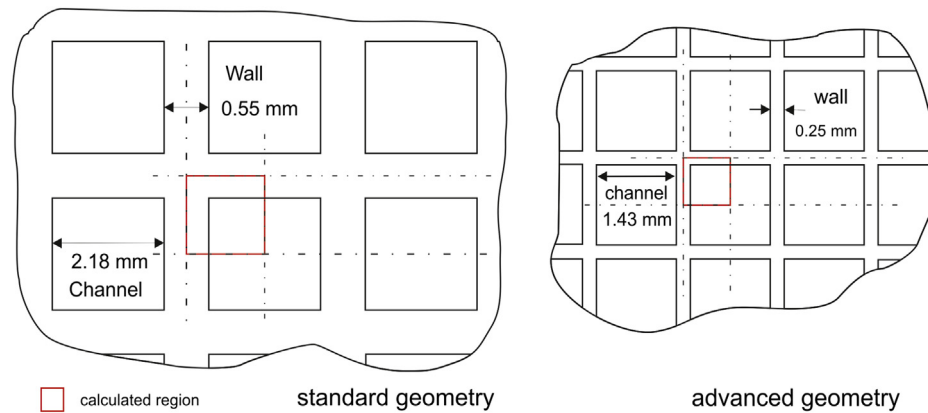


Fig. 4. Cross-sectional detail of the receiver samples.

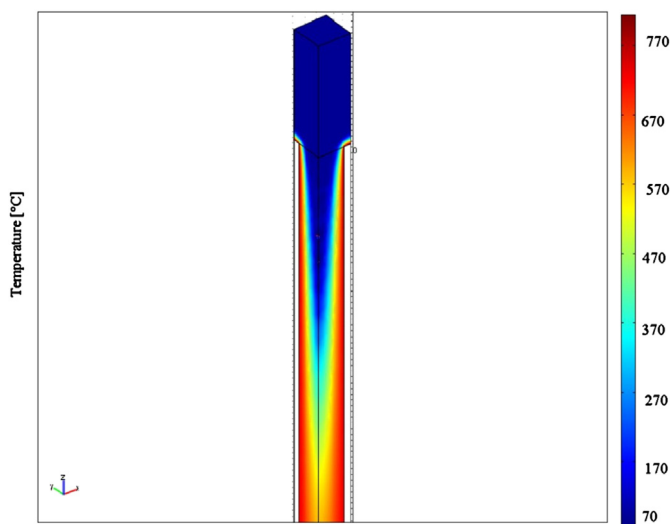


Fig. 5. Air temperature field in the channel.

solid temperature in this region. These results are confirmed by the heat flow distribution (Fig. 6), which was also used for the determination of the volumetric heat transfer coefficient as an effective quantity representing the overall heat transfer capability of the

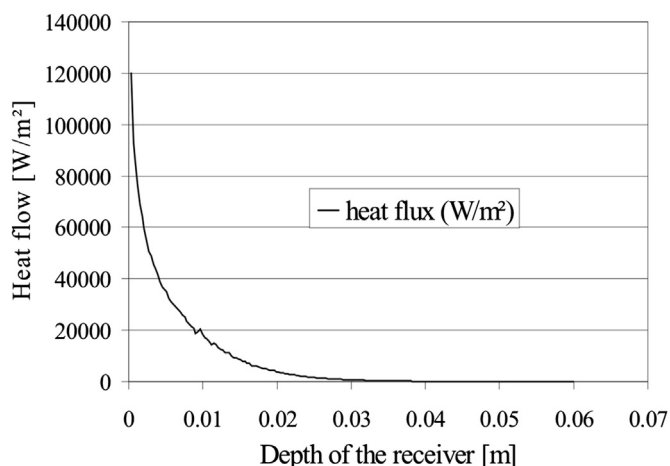


Fig. 6. Heat flow distribution in the channel.

absorber material. This quantity is later used for the continuum model.

Fig. 7 shows a comparison of the air and wall temperature in the main flow direction of the two geometries. The air temperature is taken in the center of the channel. The absorber material with smaller channels shows lower front temperatures. Furthermore the air temperature increases more quickly due to an increased surface area, which results in a lower convective heat resistance. The comparison with the experimental data is shown in Table 4.

The comparison shows that the average difference between the simulated and measured outlet temperatures and thermal efficiency is approximately 1–2%, which confirms the capability of the proposed numeric model.

3.2. The results of the continuum model

As mentioned in the Introduction, the continuum model was foreseen to investigate the influence of geometrical modifications of the whole honeycomb inlet (absorber). As modifications, an absorber sample with a frustum and one without a frustum have been compared experimentally as well as numerically using the continuum model. The geometry difference can be seen in Fig. 9. Note that due to symmetry reasons only one-quarter of the absorber has been considered, so that the white straight line and the white dashed straight line are showing the symmetry axis of cup and absorber. The motivation for choosing this frustum

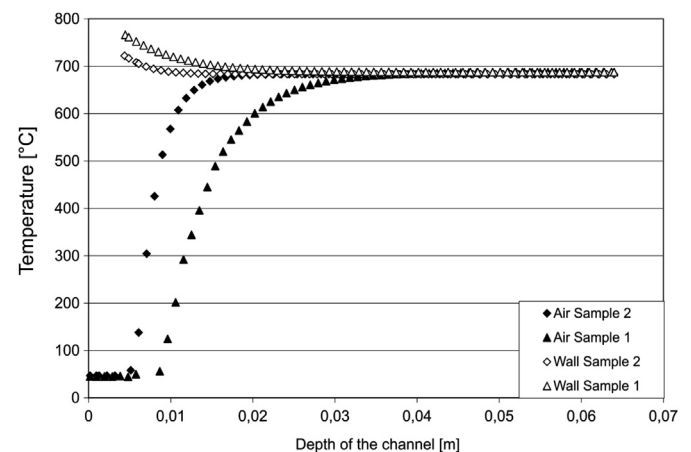
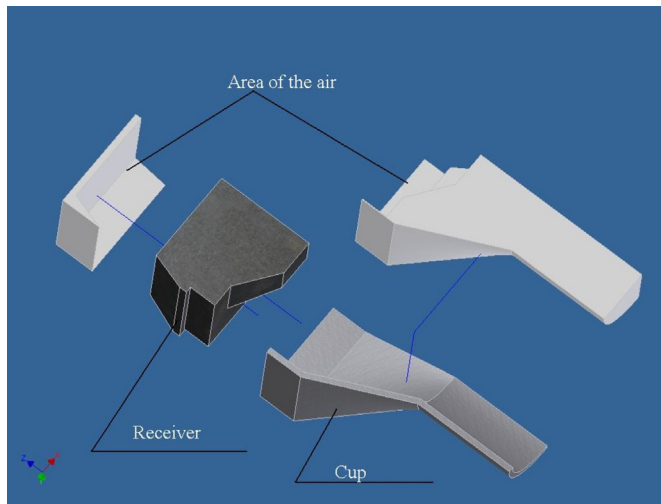


Fig. 7. Calculated air and wall temperature distribution in main flow direction (in the center of the absorber channel).

Table 4

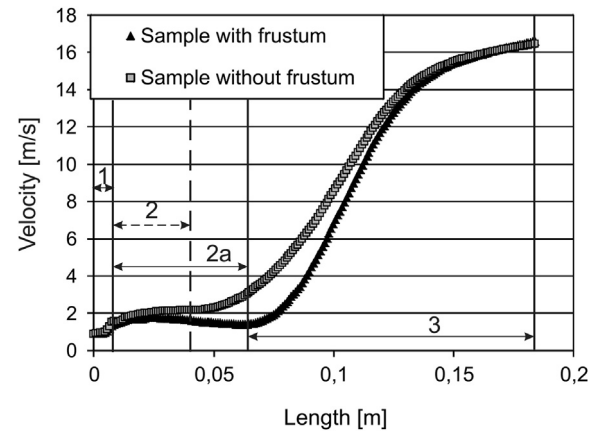
Comparison of the receiver outlet temperatures (numeric and measured).

Sample	Measured values		Simulated values	
	$\bar{T}_{out}$ [°C]	Thermal efficiency [%]	$\bar{T}_{out}$ [°C]	Thermal efficiency [%]
1	699 ± 5	72 ± 5	684	74
2	701 ± 5	75 ± 5	688	76

**Fig. 8.** The geometry of the continuum model.

geometry was the result of simple fluid mechanical considerations. Since the fluid velocity in the middle of the receiver cup was expected to be higher than at the edges, an additional pressure drop by longer channels had to be created in the central region to achieve homogeneous flow.

The irradiative flow in the experiment ranged from 300 kW/m² to 700 kW/m². To regard the experimental boundary conditions as good as possible, the integral value over the aperture area of the absorber has been calculated and divided by 4 since the volume considered was one-quarter of the absorber. This quantity has then be multiplied with the absorptivity of the absorber to use it as a volumetric heat source in the model as described in Section 2. The receiver's real geometry including absorber and containment (cup) was implemented. The version with frustum is shown in Fig. 8. The fluid domains are shown as white areas. Since the point of interest

**Fig. 10.** Comparison of the calculated (continuum model) velocity distribution in the center of the receiver (dotted line in Fig. 9).

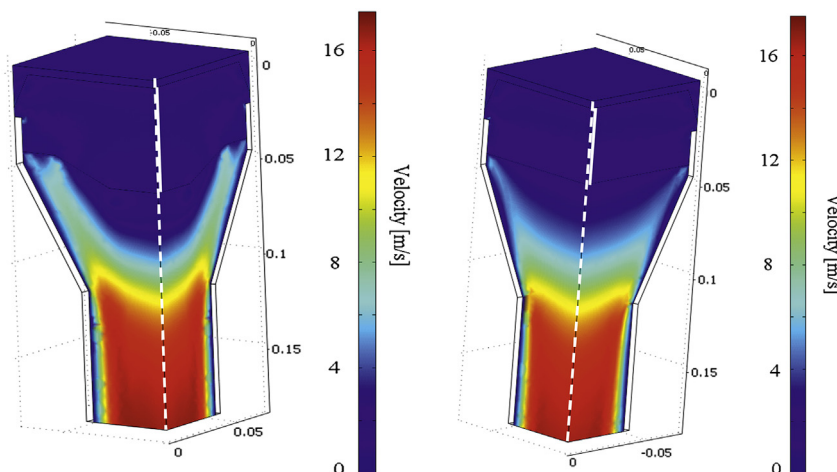
was the performance of the honeycomb material as an absorber, heat transfer at the outer surface of the cup was not regarded to reduce the model.

Fig. 9 shows the velocity distributions as a comparison between the geometry with and without frustum. The velocity field has a parabolic profile for both samples. However, gradients in the directions perpendicular to the main flow direction are significantly higher in the sample with frustum. This is due to the larger hydraulic resistance in the center of this geometry.

The diagram in Fig. 10 shows the comparison of the calculated velocity for both geometries at the center of the receiver in flow direction. The domains 1, 2, 2.1 and 3 correspond to the inlet domain, the receiver's domain without frustum, the receiver's domain with frustum and the outlet domain. The results present lower air velocities in the geometry with frustum as a consequence of the larger hydraulic resistance. In Section 3, the velocity increases due to the smaller cross-section of the cup.

Fig. 11 shows the temperature distribution at the inlet surfaces of the receiver with and without frustum. The distribution is homogeneous for the sample with frustum and non-homogeneous for the sample without frustum. The non-homogeneous distribution can be explained by the lower mass flow at the edges of the receiver element close to the cup walls in comparison to the central region.

The diagram in Fig. 12 shows the calculated solid and fluid temperature distribution along the center line of the absorber in

**Fig. 9.** Velocity fields in the two receiver geometries calculated with the continuum model.

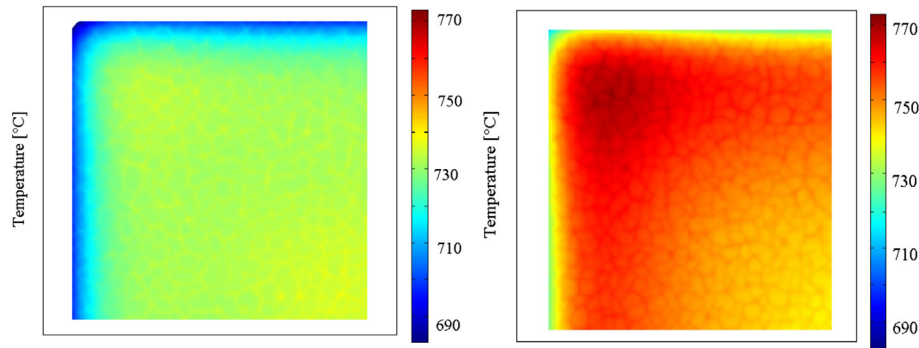


Fig. 11. Solid receiver temperatures at the inlet surface (with frustum left, without frustum right).

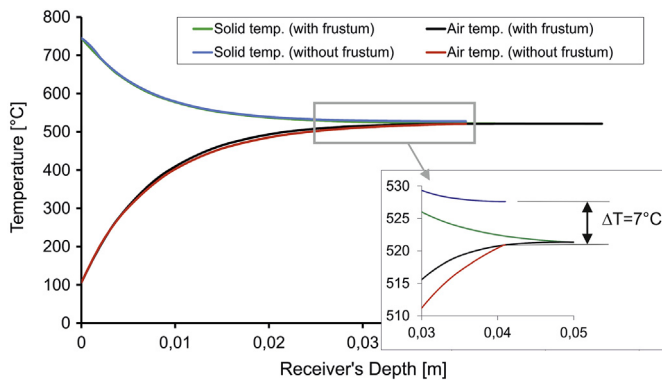


Fig. 12. The numerically determined temperature distribution in the center of the receiver volume in flow direction.

Table 5
Comparison between the experimental and numeric results of the continuum model.

	Inlet mass flow [kg/s]	$\bar{T}_{in}$ [°C]	Experimental values		Numeric values	
			$\bar{T}_{out}$ [°C]	η_{exp} [%]	$\bar{T}_{out}$ [°C]	η_{exp} [%]
Sample with frustum (1)	0.02	67 ± 2	500.5 ± 5	89.3 ± 5	517	91.7
Sample without frustum (2)	0.0189	74 ± 2	500 ± 5	82.3 ± 5	538	89.1

flow direction (continuous white line in Fig. 9). 30 mm behind the inlet in flow direction the difference between the solid and the fluid temperature without frustum is 7 °C. In case of the sample with frustum the difference between fluid and solid temperature becomes zero. The remaining difference between fluid and solid temperature in the case of the absorber without a frustum shows, that not all thermal energy of the absorber has been transferred to the fluid, thus the thermal efficiency of this sample is lower.

Table 5 gives an overview of the main experimental and numerical results. The influence of the frustum on the efficiency is positive. The correspondence between experimentally and numerically determined air outlet temperature values is good.

4. Conclusions

The comparison of the numerical and experimental results shows a correspondence for the single channel model with only 1–2% of divergence. The sample with the finer structure exhibits some advantages because of the smaller convective thermal resistance due to an increased heat transfer surface. Furthermore, the single channel model can be used to determine the volumetric convective

heat transfer coefficient of the honeycomb material investigated. This is a useful quantity, which can be used for reduced numerical models such as the continuum model.

The comparison between the geometry with and without frustum shows a larger thermal efficiency for the sample with frustum. This is true for both, the numerical as well as the experimental data reported.

Both of the presented models can be used to predict temperatures and velocity distributions in the volumetric solar air receiver. The continuum model can be further improved if the measured distribution of the concentrated solar radiation is considered in more detail. Finally it can be used as a tool to further optimize the receiver by geometric modifications.

Acknowledgments

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